

# Applying Exponent Laws

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## MA.8.NSO.1.3

*I can rewrite a rational base with a negative exponent using a positive exponent, and apply several laws to generate an equivalent expression.*

*Yesterday: a negative exponent means reciprocal.*

**Today the base is a fraction — so the reciprocal flips it.**

# Warm-Up

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*On your own. Four minutes. No notes.*

1. Find the value:  $2^{-3}$
2. Rewrite with a positive exponent:  $\frac{1}{7^{-2}}$
3. Which is larger,  $4^3$  or  $3^4$ ?
4. Write the reciprocal of  $-\frac{2}{5}$

# Warm-Up

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*Answers.*

1. Find the value:  $2^{-3}$   $\frac{1}{8}$
2. Rewrite with a positive exponent:  $\frac{1}{7^{-2}}$   $7^2$
3. Which is larger,  $4^3$  or  $3^4$ ?  $3^4$  (**81 > 64**)
4. Write the reciprocal of  $-\frac{2}{5}$   $-\frac{5}{2}$

# Notes

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*Copy the table.*

## I. The pattern with a fraction base

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| Expression   | Expanded        | Value  |
|--------------|-----------------|--------|
| $(2/5)^2$    | $2/5 \cdot 2/5$ | $4/25$ |
| $(2/5)^1$    | $2/5$           | $2/5$  |
| $(2/5)^0$    | $1$             | $1$    |
| $(2/5)^{-1}$ | $5/2$           | $5/2$  |
| $(2/5)^{-2}$ | $5/2 \cdot 5/2$ | $25/4$ |

# Notes

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## II. Flip the base, make the exponent positive

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$$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n \quad (a, b \neq 0)$$

$$\left(\frac{2}{5}\right)^{-2} = \left(\frac{5}{2}\right)^2 = \frac{25}{4} \quad \left(-\frac{2}{3}\right)^{-5} = \left(-\frac{3}{2}\right)^5$$

**The sign stays with the fraction. Flip the numbers, not the sign.**

A whole number is a fraction over 1:  $4^{-3} = \left(\frac{4}{1}\right)^{-3} = \left(\frac{1}{4}\right)^3$

# Notes

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## III. Several laws at once — the plan is still yours

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$$\left(\frac{2.5^5}{2.5^3}\right)^{-1} = (2.5^2)^{-1} = 2.5^{-2} = \left(\frac{5}{2}\right)^{-2} = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$$

Any correct order of laws gives the same value. Finish with a positive exponent.

# Example 1

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*Flip, then evaluate.*

$$\left(-\frac{7}{8}\right)^{-2}$$

**Rewrite with a positive exponent, then find the value.**

# Example 1

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*Worked.*

$$\left(-\frac{7}{8}\right)^{-2} = \left(-\frac{8}{7}\right)^2 \quad \text{flip the base; the sign stays}$$

$$\left(-\frac{8}{7}\right)^2 = \frac{64}{49} \quad \text{even power: positive}$$

**Answer: 64/49**

# Your Turn

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*Same steps, your numbers. Boards up when done.*

$$\left(-\frac{5}{2}\right)^{-3}$$

# Your Turn

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*Answer.*

$$\left(-\frac{5}{2}\right)^{-3}$$

$(-2/5)^3$  — odd power, negative

**Answer:**  $-\frac{8}{125}$

# Example 2

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*Two laws, then a flip.*

$$\left(\frac{5}{8}\right)^{-2} \cdot \left(\frac{5}{8}\right)^4$$

**Write an equivalent expression with one base and a positive exponent.**

# Example 2

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*Worked.*

$$\left(\frac{5}{8}\right)^{-2} \cdot \left(\frac{5}{8}\right)^4 = \left(\frac{5}{8}\right)^{-2+4} = \left(\frac{5}{8}\right)^2 \quad \text{product of powers}$$

$$\left(\frac{5}{8}\right)^2 = \frac{25}{64} \quad \text{already positive: no flip}$$

**Answer:  $(5/8)^2 = 25/64$**

# Your Turn

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*Same steps, your numbers. Boards up when done.*

$$\left(-\frac{5}{4}\right)^{-7} \cdot \left(-\frac{5}{4}\right)^{-3}$$

# Your Turn

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*Answer.*

$$\left(-\frac{5}{4}\right)^{-7} \cdot \left(-\frac{5}{4}\right)^{-3}$$

exponent  $-10$ , then flip:  $(-4/5)^{10}$

$$\textbf{Answer: } \left(-\frac{4}{5}\right)^{10}$$

# Whiteboards • Question 1 of 9

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*Boards up on three.*

A copier is set to shrink a page to  $\frac{2}{3}$  of its size.

A page 6 inches wide is copied 4 times in a row, each copy from the last.

**What fraction of the original page's size is the last copy?**

**Answer it.**

# Whiteboards • Question 1 of 9

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*Answer.*

A copier is set to shrink a page to  $\frac{2}{3}$  of its size.

A page 6 inches wide is copied 4 times in a row, each copy from the last.

**What fraction of the original page's size is the last copy?**

$$\left(\frac{2}{3}\right)^4 = \frac{2^4}{3^4}$$

**Answer:**  $\frac{16}{81}$

# Whiteboards • Question 2 of 9

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*Boards up on three.*

$$\left(\frac{3}{-5}\right)^{-9}$$

**Answer it.**

*Rewrite with a positive exponent.*

# Whiteboards • Question 2 of 9

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*Answer.*

$$\left(\frac{3}{-5}\right)^{-9}$$

flip the numbers; the sign stays in the fraction

**Answer:**  $\left(\frac{-5}{3}\right)^9$

# Whiteboards • Question 3 of 9

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*Boards up on three.*

$$\left(-\frac{8}{9}\right)^{-2}$$

**Answer it.**

*Find the value.*

# Whiteboards • Question 3 of 9

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*Answer.*

$$\left(-\frac{8}{9}\right)^{-2}$$

$$(-9/8)^2 = 81/64$$

$$\textbf{Answer: } \frac{81}{64}$$

# Whiteboards • Question 4 of 9

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*Boards up on three.*

$$\left(\frac{21}{4}\right)^{-5}$$

**Answer it.**

*Rewrite with a positive exponent.*

# Whiteboards • Question 4 of 9

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*Answer.*

$$\left(\frac{21}{4}\right)^{-5}$$

flip

**Answer:**  $\left(\frac{4}{21}\right)^5$

# Whiteboards • Question 5 of 9

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*Boards up on three.*

$$\left(-\frac{7}{8}\right)^{-3}$$

Which expression is equivalent?

**A.**  $(7/8)^3$

**B.**  $(-8/7)^3$

**C.**  $(-7/8)^3$

**D.**  $(8/7)^3$

**Answer it.**

# Whiteboards • Question 5 of 9

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*Answer.*

$$\left(-\frac{7}{8}\right)^{-3}$$

Which expression is equivalent?

**A.**  $(7/8)^3$

**B.**  $(-8/7)^3$

**C.**  $(-7/8)^3$

**D.**  $(8/7)^3$

**Answer: B —  $(-8/7)^3$**

# Whiteboards • Question 6 of 9

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*Boards up on three.*

$$\left(\frac{5}{8}\right)^{-2} \div \left(\frac{5}{8}\right)^{-4}$$

Which expression is equivalent?

**A.**  $(8/5)^2$

**B.**  $(5/8)^{-6}$

**C.**  $(5/8)^2$

**D.**  $(5/8)^8$

**Answer it.**

# Whiteboards • Question 6 of 9

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*Answer.*

$$\left(\frac{5}{8}\right)^{-2} \div \left(\frac{5}{8}\right)^{-4}$$

Which expression is equivalent?

**A.**  $(8/5)^2$

**B.**  $(5/8)^{-6}$

**C.**  $(5/8)^2$

**D.**  $(5/8)^8$

**Answer: C —  $(5/8)^2$**

# Whiteboards • Question 7 of 9

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*Boards up on three.*

$$\left(\frac{2}{5}\right)^{-1} \cdot \left(\frac{5}{2}\right)^1$$

**Answer it.**

*Find the value.*

# Whiteboards • Question 7 of 9

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*Answer.*

$$\left(\frac{2}{5}\right)^{-1} \cdot \left(\frac{5}{2}\right)^1$$

$$(5/2)^1 \cdot (5/2)^1 = (5/2)^2$$

$$\text{Answer: } \frac{25}{4}$$

# Whiteboards • Question 8 of 9

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*Boards up on three.*

$$\left[\left(\frac{2}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^{-2}\right]^3$$

**Answer it.**

*Write it with one base and one positive exponent.*

# Whiteboards • Question 8 of 9

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*Answer.*

$$\left[ \left( \frac{2}{3} \right)^5 \cdot \left( \frac{2}{3} \right)^{-2} \right]^3$$

$(2/3)^3$  inside, then power of a power

**Answer:**  $\left( \frac{2}{3} \right)^9$

# Whiteboards • Question 9 of 9

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*Take your time. Boards up when you have written it.*

Marco says  $\left(-\frac{2}{3}\right)^{-5}$  can be rewritten as  $\left(\frac{2}{3}\right)^5$

because 'two negatives make a positive.' Is he right?

**Write your answer in sentences.**

*Say what the negative exponent does, and what it does not do. Give the correct rewrite.*

# Whiteboards • Question 9 of 9

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*Answer.*

Marco says  $\left(-\frac{2}{3}\right)^{-5}$  can be rewritten as  $\left(\frac{2}{3}\right)^5$

because 'two negatives make a positive.' Is he right?

the exponent flips the fraction; it never touches the sign

**Answer: No —  $(-2/3)^{-5} = (-3/2)^5$**

# Independent Set

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*Six questions. On your own, in silence.*

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

# IXL

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*Last five minutes.*

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

## **Today's skills — all required**

Identify equivalent expressions involving exponents I — VLM